

Argue that if X and Y are nonnegative random variables, then $E[\max \{X, Y\}] \leq E[X] + E[Y]$.

★ C.3-5

Let X and Y be independent random variables. Prove that $f(X)$ and $g(Y)$ are independent for any choice of functions f and g .

★ C.3-6

Let X be a nonnegative random variable, and suppose that $E[X]$ is well defined. Prove *Markov's inequality*:

$$\Pr \{X \geq t\} \leq E[X] / t \quad (\text{C.34})$$

for all $t > 0$.

★ C.3-7

Let S be a sample space, and let X and X' be random variables such that $X(s) \geq X'(s)$ for all $s \in S$. Prove that for any real constant t ,

$$\Pr \{X \geq t\} \geq \Pr \{X' \geq t\}.$$

C.3-8

Which is larger: the expectation of the square of a random variable, or the square of its expectation?

C.3-9

Show that for any random variable X that takes on only the values 0 and 1, we have $\text{Var}[X] = E[X] E[1 - X]$.

C.3-10

Prove that $\text{Var}[aX] = a^2 \text{Var}[X]$ from the definition (C.31) of variance.

C.4 The geometric and binomial distributions

A **Bernoulli trial** is an experiment with only two possible outcomes: **success**, which occurs with probability p , and **failure**, which occurs with probability $q = 1 - p$. A coin flip serves as an example where, depending on your point of view, heads equates to success and tails to failure. When we speak of **Bernoulli trials** collectively, we mean that the trials are mutually independent and, unless we specifically say otherwise, that each has the same probability p for success. Two important distributions arise from Bernoulli trials: the geometric distribution and the binomial distribution.

The geometric distribution

Consider a sequence of Bernoulli trials, each with a probability p of success and a probability $q = 1 - p$ of failure. How many trials occur before a success? Define the random variable X to be the number of trials needed to obtain a success. Then X has values in the range $\{1, 2, \dots\}$, and for $k \geq 1$,

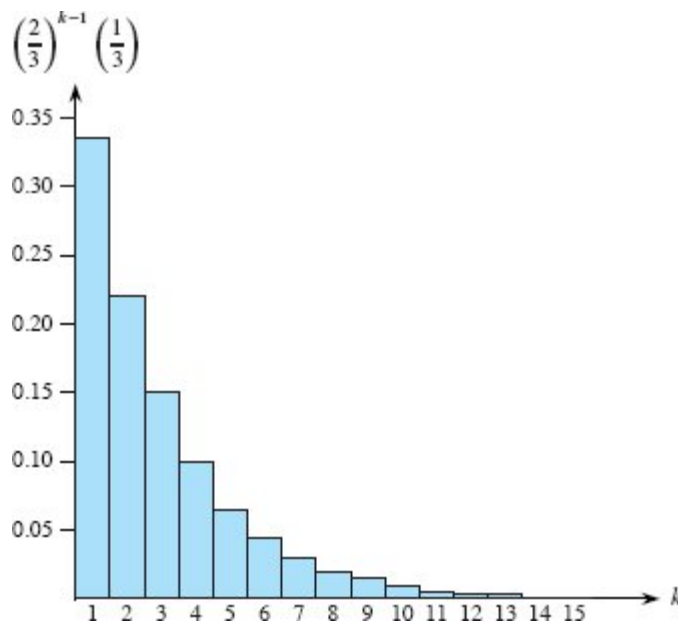


Figure C.1 A geometric distribution with probability $p = 1/3$ of success and a probability $q = 1 - p$ of failure. The expectation of the distribution is $1/p = 3$.

$$\Pr\{X = k\} = q^{k-1} p, \quad (\text{C.35})$$

since $k - 1$ failures occur before the first success. A probability distribution satisfying equation (C.35) is said to be a *geometric distribution*. Figure C.1 illustrates such a distribution.

Assuming that $q < 1$, we can calculate the expectation of a geometric distribution:

$$\begin{aligned}
 E[X] &= \sum_{k=1}^{\infty} k q^{k-1} p \\
 &= \frac{p}{q} \sum_{k=0}^{\infty} k q^k \\
 &= \frac{p}{q} \cdot \frac{q}{(1-q)^2} \quad (\text{by equation (A.11) on page 1142}) \\
 &= \frac{p}{q} \cdot \frac{q}{p^2} \\
 &= 1/p .
 \end{aligned} \tag{C.36}$$

Thus, on average, it takes $1/p$ trials before a success occurs, an intuitive result. As Exercise C.4-3 asks you to show, the variance is

$$\text{Var}[X] = q/p^2 . \tag{C.37}$$

As an example, suppose that you repeatedly roll two dice until you obtain either a seven or an eleven. Of the 36 possible outcomes, 6 yield a seven and 2 yield an eleven. Thus, the probability of success is $p = 8/36 = 2/9$, and you'd have to roll $1/p = 9/2 = 4.5$ times on average to obtain a seven or eleven.

The binomial distribution

How many successes occur during n Bernoulli trials, where a success occurs with probability p and a failure with probability $q = 1 - p$? Define the random variable X to be the number of successes in n trials. Then X has values in the range $\{0, 1, \dots, n\}$, and for $k = 0, 1, \dots, n$,

$$\Pr\{X = k\} = \binom{n}{k} p^k q^{n-k} , \tag{C.38}$$

since there are $\binom{n}{k}$ ways to pick which k of the n trials are successes, and the probability that each occurs is $p^k q^{n-k}$. A probability distribution satisfying equation (C.38) is said to be a *binomial distribution*. For

convenience, we define the family of binomial distributions using the notation

$$b(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k} . \quad (\text{C.39})$$

Figure C.2 illustrates a binomial distribution. The name “binomial” comes from the right-hand side of equation (C.38) being the k th term of the expansion of $(p + q)^n$. Consequently, since $p + q = 1$, equation (C.4) on page 1181 gives

$$\sum_{k=0}^n b(k; n, p) = 1 , \quad (\text{C.40})$$

as axiom 2 of the probability axioms requires.

We can compute the expectation of a random variable having a binomial distribution from equations (C.9) and (C.40). Let X be a random variable that follows the binomial distribution $b(k; n, p)$, and let $q = 1 - p$. The definition of expectation gives

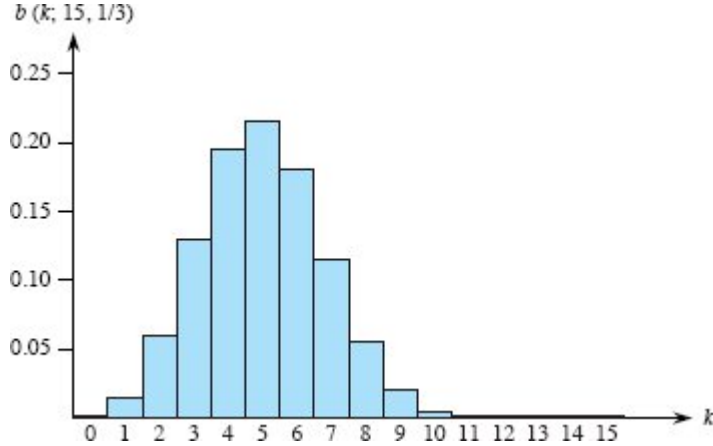


Figure C.2 The binomial distribution $b(k; 15, 1/3)$ resulting from $n = 15$ Bernoulli trials, each with probability $p = 1/3$ of success. The expectation of the distribution is $np = 5$.

$$\begin{aligned}
E[X] &= \sum_{k=0}^n k \cdot \Pr\{X = k\} \\
&= \sum_{k=0}^n k \cdot b(k; n, p) \\
&= \sum_{k=1}^n k \binom{n}{k} p^k q^{n-k} \\
&= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} q^{n-k} \quad (\text{by equation (C.9) on page 1183}) \\
&= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k q^{(n-1)-k} \\
&= np \sum_{k=0}^{n-1} b(k; n-1, p) \\
&= np \quad (\text{by equation (C.40)}) . \tag{C.41}
\end{aligned}$$

Linearity of expectation produces the same result with substantially less algebra. Let X_i be the random variable describing the number of successes in the i th trial. Then $E[X_i] = p \cdot 1 + q \cdot 0 = p$, and the expected number of successes for n trials is

$$\begin{aligned}
E[X] &= E\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n E[X_i] \quad (\text{by equation (C.24) on page 1192}) \\
&= \sum_{i=1}^n p \\
&= np . \tag{C.42}
\end{aligned}$$

We can use the same approach to calculate the variance of the distribution. By equation (C.31), $\text{Var}[X_i] = E[X_i^2] - E^2[X_i]$. Since X_i takes on only the values 0 and 1, we have $X_i^2 = X_i$, which implies $E[X_i^2] = E[X_i] = p$. Hence,

$$\text{Var}[X_i] = p - p^2 = p(1 - p) = pq . \tag{C.43}$$

To compute the variance of X , we take advantage of the independence of the n trials. By equation (C.33), we have

$$\begin{aligned}
\text{Var}[X] &= \text{Var}\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n \text{Var}[X_i] \\
&= \sum_{i=1}^n pq \\
&= npq.
\end{aligned} \tag{C.44}$$

As Figure C.2 shows, the binomial distribution $b(k; n, p)$ increases with k until it reaches the mean np , and then it decreases. To prove that the distribution always behaves in this manner, examine the ratio of successive terms:

$$\begin{aligned}
\frac{b(k; n, p)}{b(k-1; n, p)} &= \frac{\binom{n}{k} p^k q^{n-k}}{\binom{n}{k-1} p^{k-1} q^{n-k+1}} \\
&= \frac{n! (k-1)! (n-k+1)! p}{k! (n-k)! n! q} \\
&= \frac{(n-k+1)p}{kq} \\
&= 1 + \frac{(n-k+1)p - kq}{kq} \\
&= 1 + \frac{(n-k+1)p - k(1-p)}{kq} \\
&= 1 + \frac{(n+1)p - k}{kq}.
\end{aligned} \tag{C.45}$$

This ratio is greater than 1 precisely when $(n+1)p - k$ is positive. Consequently, $b(k; n, p) > b(k-1; n, p)$ for $k < (n+1)p$ (the distribution increases), and $b(k; n, p) < b(k-1; n, p)$ for $k > (n+1)p$ (the distribution decreases). If $(n+1)p$ is an integer, then for $k = (n+1)p$, the ratio $b(k; n, p)/b(k-1; n, p)$ equals 1, so that $b(k; n, p) = b(k-1; n, p)$. In this case, the distribution has two maxima: at $k = (n+1)p$ and at $k-1 = (n+1)p-1 = np-q$. Otherwise, it attains a maximum at the unique integer k that lies in the range $np - q < k < (n+1)p$.

The following lemma provides an upper bound on the binomial distribution.

Lemma C.1

Let $n \geq 0$, let $0 < p < 1$, let $q = 1 - p$, and let $0 \leq k \leq n$. Then

$$b(k; n, p) \leq \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.$$

Proof We have

$$\begin{aligned} b(k; n, p) &= \binom{n}{k} p^k q^{n-k} \\ &\leq \left(\frac{n}{k}\right)^k \left(\frac{n}{n-k}\right)^{n-k} p^k q^{n-k} \quad (\text{by inequality (C.7) on page 1182}) \\ &= \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}. \end{aligned}$$

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Exercises

C.4-1

Verify axiom 2 of the probability axioms for the geometric distribution.

C.4-2

How many times on average do you need to flip six fair coins before obtaining three heads and three tails?

C.4-3

Show that the variance of the geometric distribution is q/p^2 . (*Hint*: Use Exercise A.1-6 on page 1144.)

C.4-4

Show that $b(k; n, p) = b(n - k; n, q)$, where $q = 1 - p$.

C.4-5

Show that the value of the maximum of the binomial distribution $b(k; n, p)$ is approximately $1/\sqrt{2\pi npq}$, where $q = 1 - p$.

★ C.4-6

Show that the probability of no successes in n Bernoulli trials, each with probability $p = 1/n$ of success, is approximately $1/e$. Show that the

probability of exactly one success is also approximately $1/e$.

★ C.4-7

Professor Rosencrantz flips a fair coin n times, and so does Professor Guildenstern. Show that the probability that they get the same number of heads is $\binom{2n}{n}/4^n$. (*Hint:* For Professor Rosencrantz, call a head a success, and for Professor Guildenstern, call a tail a success.) Use your argument to verify the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

★ C.4-8

Show that for $0 \leq k \leq n$,

$$b(k; n, 1/2) \leq 2^n H(k/n) - n,$$

where $H(x)$ is the entropy function (C.8) on page 1182.

★ C.4-9

Consider n Bernoulli trials, where for $i = 1, 2, \dots, n$, the i th trial has probability p_i of success, and let X be the random variable denoting the total number of successes. Let $p \geq p_i$ for all $i = 1, 2, \dots, n$. Prove that for $1 \leq k \leq n$,

$$\Pr\{X < k\} \geq \sum_{i=0}^{k-1} b(i; n, p).$$

★ C.4-10

Let X be the random variable for the total number of successes in a set A of n Bernoulli trials, where the i th trial has a probability p_i of success, and let X' be the random variable for the total number of successes in a second set A' of n Bernoulli trials, where the i th trial has a probability $p'_i \geq p_i$ of success. Prove that for $0 \leq k \leq n$,

$$\Pr\{X' \geq k\} \geq \Pr\{X \geq k\}.$$

(Hint: Show how to obtain the Bernoulli trials in A' by an experiment involving the trials of A , and use the result of Exercise C.3-7.)

★ C.5 The tails of the binomial distribution

The probability of having at least, or at most, k successes in n Bernoulli trials, each with probability p of success, is often of more interest than the probability of having exactly k successes. In this section, we investigate the *tails* of the binomial distribution: the two regions of the distribution $b(k; n, p)$ that are far from the mean np . We'll prove several important bounds on (the sum of all terms in) a tail.

We first provide a bound on the right tail of the distribution $b(k; n, p)$. To determine bounds on the left tail, simply invert the roles of successes and failures.

Theorem C.2

Consider a sequence of n Bernoulli trials, where success occurs with probability p . Let X be the random variable denoting the total number of successes. Then for $0 \leq k \leq n$, the probability of at least k successes is

$$\begin{aligned} \Pr\{X \geq k\} &= \sum_{i=k}^n b(i; n, p) \\ &\leq \binom{n}{k} p^k. \end{aligned}$$

Proof For $S \subseteq \{1, 2, \dots, n\}$, let A_S denote the event that the i th trial is a success for every $i \in S$. Since $\Pr\{A_S\} = p^{|S|}$, where $|S| = k$, we have

$$\begin{aligned} \Pr\{X \geq k\} &= \Pr\{\text{there exists } S \subseteq \{1, 2, \dots, n\} : |S| = k \text{ and } A_S\} \\ &= \Pr\left\{\bigcup_{S \subseteq \{1, 2, \dots, n\} : |S|=k} A_S\right\} \\ &\leq \sum_{S \subseteq \{1, 2, \dots, n\} : |S|=k} \Pr\{A_S\} \quad (\text{by inequality (C.21) on page 1190}) \\ &= \binom{n}{k} p^k. \end{aligned}$$

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